

$$6(a)(i) \quad V_{\text{rms}} = \frac{V_0}{\sqrt{2}} = \frac{170}{\sqrt{2}} = 120 \text{ V} \quad [1]$$

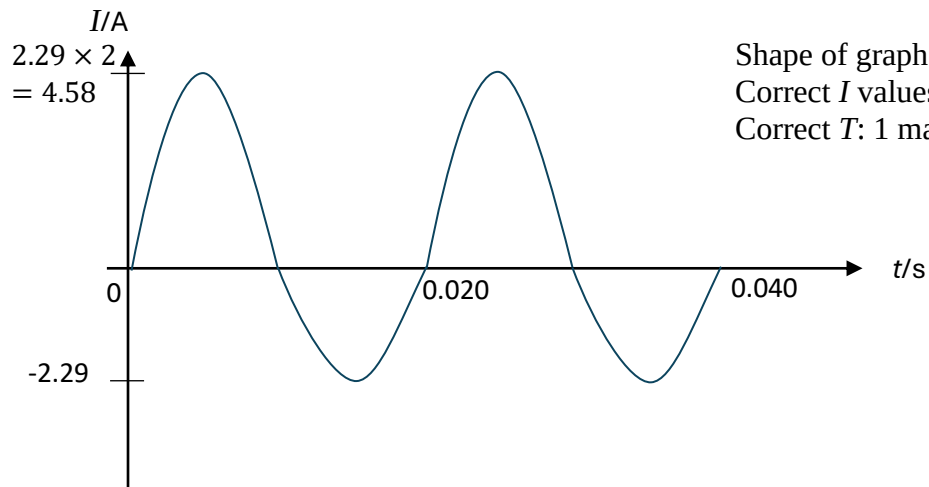
$$(ii) \quad \begin{aligned} \omega &= 2\pi/T = 314 \\ T &= 0.0200 \text{ s} \end{aligned} \quad [1]$$

(b)(i)

$$\begin{aligned} \frac{V_S}{V_P} &= \frac{N_S}{N_P} \\ \frac{V_S}{170} &= \frac{3500}{2000} \quad [1] \\ V_S &= 298 \text{ V} \quad [1] \end{aligned}$$

$$(ii) \quad \begin{aligned} V_S &= I_S R \\ 298 &= I_S (130) \\ I_S &= 2.288 \text{ A} \quad [1] \\ \frac{I_P}{I_S} &= \frac{N_S}{N_P} = \frac{3500}{2000} \\ I_P &= 4.00 \text{ A} \quad [1] \end{aligned}$$

(c)



Shape of graph: 1 mark.
Correct I values: 1 mark
Correct T : 1 mark